

Multiple Choice (10 marks)

1. A grating spectrometer can just barely resolve two wavelengths of 500 nm and 502 nm, respectively. Which of the following gives the resolving power of the spectrometer?
 - (a) 2
 - (b) 250
 - (c) 5,000
 - (d) 10,000
2. The emission spectrum of the doubly ionized lithium atom Li^{++} ($Z = 3$, $A = 7$) is identical to that of a hydrogen atom in which all the wavelengths are
 - (a) decreased by a factor of 9
 - (b) decreased by a factor of 49
 - (c) decreased by a factor of 81
 - (d) increased by a factor of 9
3. De Broglie hypothesized that the linear momentum and wavelength of a free massive particle are related by which of the following constants?
 - (a) Planck's constant
 - (b) Boltzmann's constant
 - (c) The Rydberg constant
 - (d) The speed of light
4. The best type of laser with which to do spectroscopy over a range of visible wavelengths is
 - (a) a dye laser
 - (b) a helium-neon laser
 - (c) an excimer laser
 - (d) a ruby laser
5. Which of the following gives the total spin quantum number of the electrons in the ground state of neutral nitrogen ($Z = 7$)?
 - (a) $1/2$
 - (b) 1
 - (c) $3/2$
 - (d) $5/2$

True / False (10 marks)

Answer True or False

6. True or False: The resolving power of the grating spectrometer for wavelengths of 500 nm and 502 nm is 250.
7. True or False: The magnetic susceptibility of a doped semiconductor can determine the sign of its charge carriers.
8. True or False: An electron in the $n = 4, l = 1$ state can transition to the $n = 3, l = 2$ state because it increases the angular momentum quantum number.
9. True or False: In a single-electron atom with $l = 2$, there are 5 allowed values for the magnetic quantum number m_l : -2, -1, 0, 1, and 2.
10. True or False: In a reversible thermodynamic process, the entropy of the system and its environment remains unchanged.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the total radiation force exerted on a completely reflecting surface of 3 m^2 at Earth's surface with a solar flux of $1,000 \text{ W/m}^2$. Discuss the underlying principles involved.
12. Discuss the effects of doubling the frequency of laser light on the separation of bright fringes observed in a double-slit experiment with a slit separation of 0.5 micrometers.

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